

课堂笔记整理4.14

例7-13

$$(1) t < 0 \text{ 时}, i(0^-) = i(0) = i(0^+) = 0$$

$$(2) 0 \leq t \leq 4 \text{ 时}, i(\infty) = \frac{40}{4+6} = 4 \text{ A}, R_{th} = 4+6 = 10 \Omega$$

$$\tau = \frac{L}{R_{th}} = \frac{5}{10} = \frac{1}{2} \text{ s}$$

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}} = 4 + (0-4)e^{-2t}$$

$$= 4(1 - e^{-2t}) \text{ A}, 0 \leq t \leq 4$$

$$i(4) = i(4^-) = 4(1 - e^{-8}) \approx 4 \text{ A}$$

$$(3) \text{ 当 } t \geq 4 \text{ 时, 设 P 点电压为 } V, i_1 + i_2 = i \Rightarrow \frac{40-V}{4} + \frac{10-V}{2} = \frac{V}{6} \Rightarrow V = \frac{180}{11} \text{ V}$$

$$i(\infty) = \frac{V}{6} = \frac{30}{11} = 2.727 \text{ A}, R_{th} = 4 \parallel 2 + 6 = \frac{22}{3} \Omega$$

$$\tau = \frac{L}{R_{th}} = \frac{5}{\frac{22}{3}} = \frac{15}{22} \text{ s}, i(t) = i(\infty) + [i(4) - i(\infty)]e^{-\frac{(t-4)}{\tau}}$$

$$= 2.727 + (4 - 2.727)e^{-\frac{t-4}{\tau}}$$

$$= 2.727 + 1.273e^{-1.4667(t-4)} \quad t \geq 4$$

$$i(t) = \begin{cases} 0, & t \leq 0 \\ 4(1 - e^{-2t}), & 0 \leq t \leq 4 \\ 2.727 + 1.273e^{-1.4667(t-4)}, & t \geq 4 \end{cases}$$

$$t = 2 \text{ 时}, i(2) = 4(1 - e^{-4}) = 3.93 \text{ A}$$

$$t = 5 \text{ 时}, i(5) = 2.727 + 1.273e^{-1.4667} = 3.02 \text{ A}$$

练8-1

$$(a) t < 0 \text{ 时}, i(0^-) = \frac{24}{10+2} = 2 \text{ A}, v(0^-) = 2 \times 2 = 4 \text{ V}$$

$$\text{则 } i(0^+) = i(0^-) = 2 \text{ A}, v(0^+) = v(0^-) = 4 \text{ V}$$

$$(b) \text{ 开合后 } \begin{cases} v_L = 24 - v \\ 0.4 \frac{di}{dt} = 24 - v \end{cases} \quad \text{代入 } v = 4, \text{ 得 } \frac{di(0^+)}{dt} = 50 \text{ A/s}$$

$$\text{又由 } i = \frac{v}{Z} + C \frac{dv}{dt}, C = \frac{1}{20} = 0.05 \text{ F}, \text{ 代入 } i(0^+) = 2, v(0^+) = 4,$$

$$\text{得 } 2 = 2 + 0.05 \frac{dv(0^+)}{dt}, \text{ 得 } \frac{dv(0^+)}{dt} = 0$$

$$(c) i(\infty) = \frac{24}{2} = 12 \text{ A}$$

$$v(\infty) = 2 \times 12 = 24 \text{ V}$$